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Recombination coefficients of O and N radicals on stainless steel

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Surface recombination coefficients of O and N radicals in pure O₂ and N₂ plasmas, respectively, have been estimated on the stainless steel walls of a low-pressure inductively coupled plasma reactor. The recombination coefficients are estimated using a steady state plasma model describing the balance between the volume generation of the radicals from electron-impact dissociation of the parent molecules, and the loss of the radicals due to surface recombination. The model uses radical and parent molecule number densities and the electron energy distribution function (EEDF) as input parameters. We have measured the radical number density using appearance potential mass spectrometry. The parent neutral number density is measured using mass spectrometry. The EEDF is measured using a Langmuir probe. The recombination coefficient of O radicals on stainless steel walls at approximately 330 K is estimated to be 0.17 ± 0.02 , and agrees well with previous measurements. The recombination coefficient of N radicals is estimated to be 0.07 ± 0.02 on stainless steel at 330 K. © 2000 American Institute of Physics. [S0021-8979(00)00519-3]

I. INTRODUCTION

Oxygen and nitrogen discharges are frequently used in the microelectronics industry, both as pure feed gases and as additives to other gases. Inductively coupled plasma (ICP) sources have recently become prevalent for generating low-pressure high-density plasmas. The inductively coupled radio frequency (rf) discharges with rf-biased substrates can provide relatively high-density plasma with independent control of plasma density and ion energy. These tools are typically operated at pressures of 1–50 mTorr and applied powers of 200–2000 W with the plasma densities in the range of 10^{10} – 10^{12} cm⁻³. Recently, there has been an interest in better understanding the chemistry in ICP sources. Spatially averaged global models have developed for an oxygen plasma.^{1–3} Lee *et al.*² have shown that fractional dissociation of the gas in O₂ and Cl₂ plasmas depends strongly on the wall recombination coefficient, which was treated as a parameter in their plasma model. The wall recombination coefficient of the radicals influences the neutral and ionic composition of the plasma, especially in the low-pressure plasmas, as the volume recombination reaction rates are often negligible.

Meyerson⁴ and Greaves *et al.*⁵ have measured recombination coefficients of O radicals on various metals in the 0.005–0.24 range. These measurements were done at a pressure of few Torr. However, the measurements were not performed on stainless steel. Granier *et al.*⁶ estimated a recombination coefficient of O radicals in the 0.1–1 range from actinometry measurements in a downstream diffusion plasma. Booth *et al.*⁷ estimated a recombination coefficient of O radicals to be approximately 0.5 on stainless steel surfaces using actinometry in a modulated plasma. Vorobev *et al.*⁸ measured the recombination coefficient of N radicals on various metals and metal-oxides in the 10^{-6} – 10^{-1} range

at 1–12 Torr pressure using a coaxial flow reactor. Gordiets *et al.*⁹ have measured N radical recombination coefficient of 2×10^{-4} on Pyrex.

The measurements of O radicals on stainless steel by Booth *et al.*⁷ were performed in the afterglow of a pulsed plasma. The measurements are not affected by the ion bombardment on the walls. Our estimates of the recombination coefficients of O and N radicals are measured from neutral mass spectrometry and electron energy distribution function (EEDF) measurements under steady state conditions in a plasma (i.e., with ion bombardment on the stainless steel surface). A steady state model for the radical concentration in the plasma given by the balance of radical generation by electron impact and loss by wall recombination is used in conjunction with the above measurements to estimate the recombination coefficients.

II. EXPERIMENT

A. Inductively coupled plasma

The experimental system is shown in Fig. 1, and has been described in detail in our previous work.¹⁰ Briefly, an ICP is produced in a 20 cm diam metal-gasketed ultrahigh vacuum compatible stainless steel vessel. A three-turn stove-top coil is used to couple 13.56 MHz rf power into the plasma through a 9.5 mm thick alumina dielectric. The plasma is confined to 10 cm axial length by a grounded stainless steel plate. The plasma chamber has been modified slightly from the previous design¹⁰ to ensure the plasma volume is cylindrical. The bottom grounded stainless steel plate size has been increased to 19.5 cm diameter. Similarly, a 19.5 cm outer diameter annular stainless steel ring has been attached to the alumina dielectric holder. These changes ensure the plasma is restricted to a cylindrical volume, and the plasma does not diffuse past the bottom and top surfaces. The temperature of the radial stainless steel walls of the plasma reactor is measured using a thermocouple, and varied

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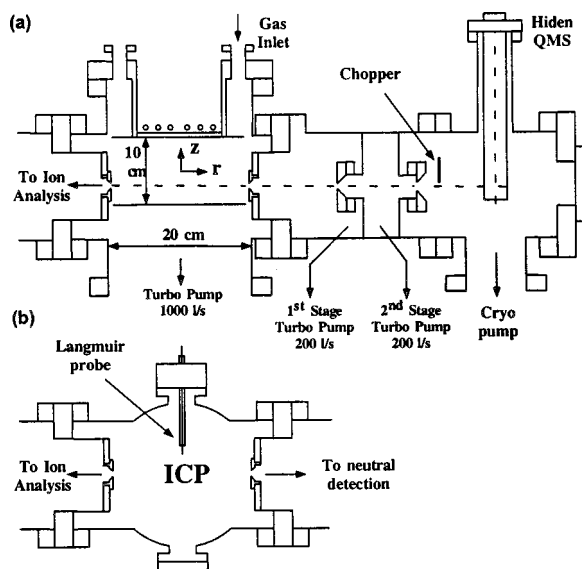


FIG. 1. Schematic diagram of the experimental system, (a) sectional front view, and (b) sectional top view.

between 320 and 340 K over the conditions studied in this article. The temperature of the bottom stainless steel plate is not measured. However, the temperature of the bottom plate is estimated from a steady state heat balance of energy input from the ion flux to the plate and heat conduction losses to the plate support. The estimated temperature of the bottom plate typically lies between 320 and 340 K for the conditions studied in this article.

B. Neutral mass spectrometry

A Hiden Analytical PIC 300 quadrupole mass spectrometer (QMS) is mounted in a cross-beam mode to detect the neutral species from the plasma. Appearance potential mass spectrometry (APMS) is used to selectively ionize and detect the radicals below the dissociative ionization threshold of the parent species. The neutrals are sampled from a 900 μm diameter aperture located 10 cm from the radial center of the plasma, and 7.6 cm below the alumina dielectric, for line-of-sight detection by the triply differentially pumped QMS. The mass spectrometer is differentially pumped by 200 l/s turbomolecular pumps in the first two stages, and a 1500 l/s cryogenic pump on the third stage. The neutral flow from the sampling aperture is in the molecular flow regime. The neutral beam emanating from the aperture reaches the mass spectrometer without suffering any gas phase or surface collisions. Modulated beam detection of the neutrals is implemented using a slotted wheel rotary beam chopper in the third differential stage to achieve background signal subtraction. This is essential even for the radical species, as they can have significant background in the mass spectrometer chamber.¹⁰ The calibration of the measured APMS signal to absolute radical number density has been discussed in detail previously.¹¹ The appearance signal of the radicals is converted to an absolute number density by calibration with an argon signal for a known argon number density in the plasma chamber measured under plasma-off condition. Calibration of the radical signal by the signal from dissociative ioniza-

tion of the parent under plasma-off conditions is avoided to eliminate the effect of discrimination against the hot fragment products from the dissociation process in the mass spectrometer ionizer.¹¹

C. Langmuir probe

A Langmuir probe is introduced in the plasma through a differentially pumped radial motion feedthrough in a plane 7.6 cm ($z = -2.5$ cm) below the dielectric as shown in Fig. 1. The Langmuir probe and the probe circuit have been described in detail previously.¹² Briefly, the probe uses a 500 μm diameter, 4 mm long cylindrical platinum tip. A copper plug held inside the alumina probe holder provides approximately 60 pF capacitance between the probe and the plasma. This reduces the rf distortion of the probe scan as the capacitance between the probe and the plasma is much larger than the capacitance between the probe and ground.¹³ Further, three tuned parallel inductive capacitive (LC) circuits, tuned to 13.56 MHz and higher harmonics, minimize the rf distortion of the probe trace by allowing the probe to follow rf voltage fluctuations in the plasma.¹³ The probe scans are obtained using a power supply interfaced to the computer through a 12 bit analog-to-digital (A/D) data acquisition board. The bias voltage is generated by $10\times$ amplification of a signal from the A/D board by the power amplifier. The current drawn by the probe is measured as a potential drop across a sensing resistor, which is isolated from the bias voltage using an isolation amplifier, and measured using the A/D board. The probe is cleaned in electron saturation until it glows red and is maintained red hot during the period of the scan to keep the probe surface clean without changing the work function of the probe surface.¹²

The electron energy probability function (EEPF) is obtained from the second derivative of the Langmuir probe characteristic in the electron retardation regime ($V_p \leq \Phi_p$, where V_p is the probe potential and Φ_p is the plasma potential)¹⁴ as follows:

$$f_0(\epsilon) \propto \frac{d^2 I_p}{dU_p^2}. \quad (1)$$

Here, $f_0(\epsilon)$ is the EEPF, I_p is the probe current and $U_p = \Phi_p - V_p$ is the probe potential referenced to the plasma potential. The zero crossing of second derivative of probe current with respect to probe potential is assumed to be the dc value of the plasma potential. The kinetic energy ϵ in Eq. (1) is simply equal to U_p . The electron energy distribution function (EEDF) is given by

$$F_0(\epsilon) = \epsilon^{1/2} f_0(\epsilon). \quad (2)$$

The plasma density is calculated from the integration of the EEDF over the entire energy range. The energy difference (Δ) between the peak and the zero crossing of I_p'' is a sensitive indicator of the overall quality and resolution of the measured EEDF.^{15,16} rf interference, probe contamination, and significant perturbation of the plasma by the probe lead to the suppression of the peak in I_p'' and the widening of Δ . In our measurements, Δ is typically about 1 eV, which helps demonstrate the accuracy of the measurements. Probe traces

TABLE I. Relative contribution of various dissociation channels to dissociation of molecular species in an oxygen plasma. The fractional contribution of a dissociation channel is given by the ratio of the rate of the channel to the sum of the rates of all the channels reported in the table. The fractional contribution of the reactions is calculated for a plasma with a 3 eV electron temperature Maxwellian EEDF and 10^{-4} fractional degree of ionization.

Reaction No.	Reaction	Rate coefficient (cm^3s^{-1})	Fractional contribution	Reference
(1)	$e + \text{O}_2 \rightarrow \text{O}(^3P) + \text{O}(^1D) + e$	$k_1 = 5.0 \times 10^{-8} e^{(-8.4/T_e)}$	0.818	2
(2)	$e + \text{O}_2 \rightarrow \text{O}(^3P) + \text{O}(^3P) + e$	$k_2 = 4.2 \times 10^{-9} e^{(-5.6/T_e)}$	0.175	3
(3)	$e + \text{O}_2 \rightarrow \text{O}(^3P) + \text{O}^- + e$	$k_3 = 8.8 \times 10^{-11} e^{(-4.4/T_e)}$	0.006	3
(4)	$e + \text{O}_2 \rightarrow \text{O} + \text{O}^+ + 2e$	$k_4 = 5.3 \times 10^{-10} T_e^{0.9} e^{(-20/T_e)}$	4.9×10^{-4}	3
(5)	$e + \text{O}_2 \rightarrow \text{O}^- + \text{O}^+ + e$	$k_5 = 7.1 \times 10^{-11} T_e^{0.5} e^{(-17/T_e)}$	1.2×10^{-4}	3
(6)	$e + \text{O}_2^+ \rightarrow \text{O} + \text{O}$	$k_6 = 5.2 \times 10^{-9}/T_e$	4.7×10^{-5}	3

are typically smoothed using five-point fast Fourier transform smoothing before numerical differentiation.

III. MODEL

The aim of this work is the determination of the recombination coefficients for O and N radicals on the stainless walls of the plasma reactor in an oxygen and a nitrogen plasma, respectively. The neutral composition of the plasma and the EEDF measurements in conjunction with a steady state model are used to calculate the recombination coefficients. The balance between the volume generation rate of the radicals and the total loss rate of the radicals gives the steady state model for the concentration of the radical (O or N), and is used to calculate the recombination coefficient of the radicals on the walls of the reactor.

A. Radical generation mechanism

It is assumed in our model that the dominant generation mechanism for the radicals is through electron impact dissociation of the ground state parent molecule at the low pressures studied in this work (10–30 mTorr). Table I gives the relative importance of various reactions that lead to dissociation of molecular species in an oxygen discharge. The relative importance of a reaction is given by the ratio of the rate of the reaction to the sum of the rates of all the reactions listed in Table I for a Maxwellian EEDF with a electron temperature (T_e) of 3 eV. Based on the fractional contribution of the different reactions listed in Table I, it is evident that the majority of dissociation (>99%) occurs through the 8.4 and 6.0 eV loss pathways [reactions (1) and (2) in Table I, respectively]. The rates of dissociative attachment, dissociative ionization, and polar dissociation of ground state O_2 [reactions (3), (4), and (5), respectively] are significantly smaller than the 6.0 and 8.4 eV dissociation pathways. Dissociation of ionic species is also considered as a possible channel for dissociation, and is found to be insignificant since the fractional degree of ionization in the plasma is of the order of 10^{-4} .

The $\text{O}_2(a^1\Delta_g)$ and $\text{O}_2(b^1\Sigma_g)$ metastables have been detected in dc discharges in the 1–10 Torr pressure range. The $\text{O}_2(a^1\Delta_g)$ is generally the dominant metastable with a concentration of up to 10% of the ground state O_2 in these discharges. The dissociation cross section for the metastable states are typically assumed to be equal to the cross section

for ground state dissociation, with the threshold reduced by the energy difference between the metastable and the ground state. The $\text{O}_2(a^1\Delta_g)$ metastable lies 0.98 eV above the ground state, and consequently, the rate constant for metastable dissociation is about 40% larger than that for the ground state for an electron temperature of 3 eV. Thus, the sum of dissociation rates of O_2 and $\text{O}_2(a^1\Delta_g)$ in a plasma with a 10% abundance of the metastable is only about 4% higher than the case in which the enhancement of the dissociation of the metastables with respect to the ground state is ignored (i.e., rate of dissociation of the metastables is taken to be equal to that of the ground state). Also, we do not find any evidence of the metastables in the mass spectrometer signal. Based on this argument, the metastables can be lumped together with ground state O_2 for the evaluation of the dissociation rate in this model. The vibrational excitation of O_2 is usually not considered to play an important role in the chemistry of O_2 discharges. Vibrational temperatures in the range of 600–800 K have been measured in dc discharges.¹⁷ Further, the total electron-impact dissociation cross section of O_2 shows little dependence on the initial vibrational state of the molecule.¹⁸ Thus, the vibrational excitation of O_2 is expected to play no significant role in the dissociation processes.

The considerations previously discussed are also valid for a low-pressure N_2 discharge. However, the role of vibrationally excited species in the dissociation processes needs to be examined in more detail for a N_2 plasma. It is well known that vibrationally excited species can play a dominant role in dc N_2 discharges in the 1–10 Torr pressure range. Vibrational temperatures are typically in the 4000–6000 K, range for these plasmas. Loureiro and Ferreira¹⁹ have shown that at low values of the reduced electric field ($E/n < 10^{-15} \text{ V cm}^2$), the vibration–vibration ($V-V$) and vibration–translation ($V-T$) energy exchange processes are the dominant pathways for dissociation in N_2 discharges. However, for $E/n > 2 \times 10^{-15} \text{ V cm}^2$, the electron-impact dissociation of N_2 becomes an equally important pathway for N_2 dissociation.¹⁹ The electron temperature in these higher pressure dc discharges are typically below 1.5 eV. The low-pressure high-density ICP used in this study typically operate at electron temperatures in 3–5 eV range (corresponding to the relatively high value of E/n , $E/n > 10^{14} \text{ V/cm}^2$). Since the electron impact processes have Arrhenius-type depen-

dence on T_e (or E/n), and the threshold of N_2 dissociation is about 10 eV, the electron-impact dissociation of N_2 is about a factor of 30 larger than the $V-V$ and $V-T$ dissociation pathways in an ICP just based on T_e considerations. Further, the rates of $V-V$ and $V-T$ processes scale as the square of the neutral number density while the rates of electron impact dissociation processes scale linearly with neutral number density. The pressures used in the study are in the 10–30 mTorr range, and, thus, the number density in our discharge is about two orders of magnitude lower than those typically used in the dc plasma studies. As a result, we estimate that the rates of $V-V$ and $V-T$ dissociation rates are about two orders of magnitudes smaller than electron impact processes based on neutral number density considerations alone. It is clear that due to the simultaneous high $T_e(E/n)$ and low neutral number density in our ICP, the electron impact dissociation of the N_2 is the dominant dissociation process. The contribution of $V-V$ and $V-T$ processes to the dissociation is about three to four orders of magnitude smaller than the contribution of electron impact dissociation processes, and is consequently neglected in our model.

It is evident from this discussion that electron impact dissociation of the parent molecules is the dominant pathway for dissociation in pure O_2 and N_2 plasmas in our ICP. We neglect the role of metastables and other excited species in the dissociation, and treat all species to be in their ground state. The uncertainties introduced due to these simplifications are estimated to be below 5%. The rate of radical generation G ($\text{cm}^{-3}\text{s}^{-1}$) at any position in the plasma is thus given by

$$G(r, z) = 2k_{\text{diss}}n_en_{M_2}, \quad (3)$$

where k_{diss} is the dissociation rate constant (cm^3s^{-1}) described in Sec. III C, and n_e and n_{M_2} are the electron and parent molecule number density (cm^3s^{-1}) at the given location in the plasma, respectively.

B. Radical loss mechanism

It is well known that the dominant loss mechanism of the radicals is wall recombination at pressures below a few Torr.¹⁷ Since our plasma operates in the 10–30 mTorr range, this assumption is exceptionally good in our case, and all volume chemistry for recombination of the radicals can be neglected. Further, the loss rate of the radicals due to flow to the pump is also negligible compared to wall loss rates. The loss rate L ($\text{cm}^{-2}\text{s}^{-1}$) of radicals due to recombination on the walls is given by²⁰

$$L = \frac{1}{4}n_M^{\text{wall}}\bar{v}_M\left(\frac{\gamma}{1-0.5\gamma}\right), \quad (4)$$

where n_M^{wall} is the number density of the radical at the wall, $\bar{v}_M$ is the thermal velocity of the radical, and γ is the recombination coefficient of the radical. The radical concentration at all locations on the walls is assumed to be equal in our model. The recombination coefficient for O and N radicals are below about 10^{-3} on alumina surface.^{5,8} Consequently, the loss of the radicals to the alumina dielectric surface is

neglected, since it is negligible compared to the loss rate of the radicals on the stainless steel surfaces in the reactor.

C. Steady state balance for the radicals

The balance of volume generation and surface loss of the radicals determines the steady state radical concentration. This can be expressed as follows:

$$\int_{-H/2}^{H/2} \int_0^R G(r, z) 2\pi r dr dz = A^{\text{wall}} L, \quad (5)$$

where $G(r, z)$ and L are the generation and loss rates defined in Eqs. (3) and (4), respectively. A^{wall} is the area of the stainless steel walls. The volume integration of $G(r, z)$ in Eq. (5) is performed over the plasma volume of radius R and axial length H . The volume generation rate of the radicals from electron impact dissociation of parent molecule of course depends on the EEDF in Eq. (3). The dependence of the rate on the EEDF in Eq. (3) arises from the dissociation rate constant k_{diss} , which is given by the convolution of the dissociation cross section $\sigma_{\text{diss}}(\epsilon)$ over the EEDF,

$$k_{\text{diss}}(r, z) = \frac{1}{n_e(r, z)} \sqrt{\frac{2e}{m}} \int_0^\infty \epsilon \sigma_{\text{diss}}(\epsilon) f_0(\epsilon, r, z) d\epsilon. \quad (6)$$

We use cross sections from Eliasson *et al.*²¹ and Cosby²² for oxygen and nitrogen dissociation, respectively. The dissociation rate constant varies over the plasma volume due to the variation in the EEDF. Therefore, to determine the total rate of generation of the radicals in the plasma, the variation of the EEDF needs to be accounted for in the model. As described in Sec. II C, we can measure the radially resolved EEDF in the plasma at $z = -2.5$ cm. These measurements are used to calculate the variation of k_{diss} over the plasma volume. Substituting Eqs. (3) and (4) in Eq. (5), and assuming that the concentration of the radicals is the same at all wall locations, yields the following relation:

$$\begin{aligned} \int_{-H/2}^{H/2} \int_0^R 2k_{\text{diss}}(r, z)n_e(r, z)n_{M_2}(r, z)2\pi r dr dz \\ = \frac{1}{4}n_M^{\text{wall}}\bar{v}_MA^{\text{wall}}\left(\frac{\gamma}{1-0.5\gamma}\right). \end{aligned} \quad (7)$$

The unknown wall recombination coefficient of the radicals is estimated from Eq. (7) using mass spectrometry and Langmuir probe measurements discussed in the next section.

IV. RESULTS AND DISCUSSION

The radical number densities in pure O_2 and N_2 discharges measured using APMS are given in Table II. The parent number densities at the aperture reported in Table II are measured by comparing the plasma-on to the plasma-off signal for the parent ionization at 70 eV electron energy.¹¹ The neutral temperature at the aperture (Table II) is calculated from the ideal gas law using the measured pressure and the total neutral number density obtained from the sum of the radical and the parent number density.¹¹

The calculation of the recombination coefficient from Eq. (7) requires the determination of $n_{M_2}(r, z)$, the parent

TABLE II. Number densities of the radicals and the parent molecules, and the neutral temperature (T_n) in pure O_2 and N_2 plasmas measured at the neutral sampling aperture using mass spectrometry. The electron number density and temperature are measured at the radial center using the Langmuir probe.

Plasma conditions (mTorr, W)	n_M (cm^{-3})	n_{M_2} (cm^{-3})	T_n (K)	n_e (cm^{-3})	T_e (V)
Oxygen plasma					
15,650	2.48×10^{13}	3.75×10^{14}	395	3.31×10^{10}	3.192
20,550	2.52×10^{13}	5.05×10^{14}	396	2.61×10^{10}	2.778
20,680	3.32×10^{13}	4.33×10^{14}	451	3.93×10^{10}	3.063
25,700	4.61×10^{13}	5.14×10^{14}	469	6.04×10^{10}	2.936
30,700	5.62×10^{13}	5.67×10^{14}	506	7.21×10^{10}	2.806
Nitrogen plasma					
15,550	7.18×10^{12}	4.75×10^{14}	327	5.0×10^{10}	3.071
15,700	8.61×10^{12}	4.30×10^{14}	360	6.76×10^{10}	3.204
20,600	8.37×10^{12}	5.36×10^{14}	387	6.41×10^{10}	2.748
30,650	1.06×10^{13}	8.42×10^{14}	370	7.76×10^{10}	2.497

neutral number density in the volume. However, the mass spectrometric determination of the parent number density is only feasible at the neutral sampling aperture in our system. The variation of the number density of the parent molecule in the plasma volume can be estimated from a calculation of the radical number density spatial profile. The radical number density is expected to peak in the plasma center and decrease near the walls due to diffusion and wall recombination. For isothermal neutrals in the plasma, the total number density is approximately constant in the plasma volume, as the pressure gradients are small. The diffusional flux of the radicals from the plasma volume is equal to the loss rate of the radicals due to recombination on the walls, as shown in the following equation:

$$\frac{D_{\text{eff}}}{\Lambda} (n_M^{\text{bulk}} - n_M^{\text{wall}}) = \frac{1}{4} n_M^{\text{wall}} \bar{v}_M \left(\frac{\gamma}{1 - 0.5\gamma} \right), \quad (8)$$

where D_{eff} is the effective diffusion coefficient of the radical, n_M^{bulk} is the bulk concentration of the radical, and Λ is the effective diffusion length given by¹

$$\frac{1}{\Lambda^2} = \left(\frac{\pi}{H} \right)^2 + \left(\frac{2.405}{R} \right)^2. \quad (9)$$

For the plasma conditions given in Table II, it can be shown from Eq. (8) that $n_M^{\text{bulk}} \leq 2n_M^{\text{wall}}$ for $\gamma < 0.2$. Figure 2 shows the approximate number density profiles of the radical and the parent molecule under these conditions. It is clear from Fig. 2 that the change in the number density of the parent molecule over the plasma volume is small compared to the magnitude of the parent number density. Thus, the number density of the parent molecule at the wall provides a good estimate for the volume number density of the parent molecule, i.e., $n_{M_2}(r, z) \approx n_{M_2}^{\text{wall}}$.

The radial distribution of the EEPF in an oxygen plasma at 25 mTorr and 700 W is shown in Fig. 3. Similarly, the radial distribution of the EEPF in a nitrogen plasma at 30 mTorr and 650 W is shown in Fig. 4. The measured EEPF at the radial positions in the plasma are plotted versus the total energy (kinetic plus potential energy) of the electrons. The potential energy of the electrons at a point away from the center is given by the difference in the plasma potential at

the center of the discharge and the point of interest. The EEPF measured using the Langmuir probe gives the kinetic part of the electron energy distribution at any given point. As seen from Figs. 3 and 4, the EEPFs at all the radial locations can be described by a single distribution function of the total electron energy. This corresponds to nonlocal electrons in a plasma, where the electron energy relaxation length is much greater than the discharge dimensions, and the EEPF can be considered as a spatially homogeneous function of the total energy of the electrons.²³ Kortshagen²⁴ has shown the applicability of nonlocal theory in an argon discharge for low pressures ($pR < 0.2$ Torr cm, where p is the pressure in the discharge). Nonlocality of the electrons in the plasma causes the plasma density to peak at the geometric center of the plasma, and leads to a symmetric plasma density profile.²⁴ It should be noted that Mümken²⁵ has shown that even when most electrons are nearly nonlocal in a plasma, the ionization rate profile in the plasma can be asymmetric and peak near

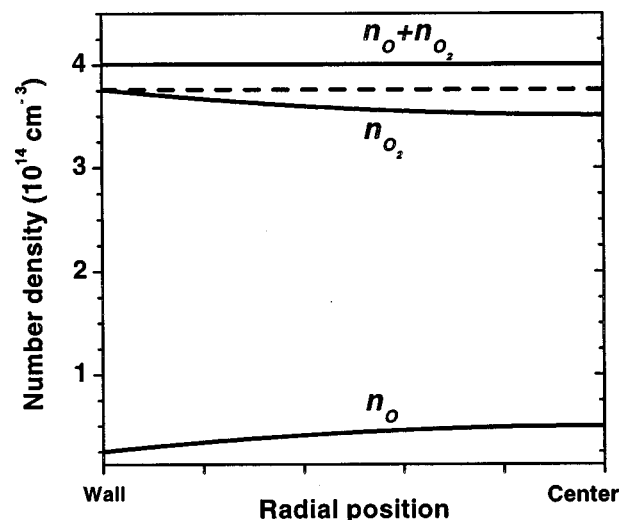


FIG. 2. Estimate of the variation of number densities with respect to either radial (or axial) position. The wall number densities shown correspond to 10 mTorr, 650 W oxygen plasma. The approximate number density profiles for O and O_2 are shown. The dashed line corresponds to the assumption that O_2 number density in the volume is equal to that at the wall.

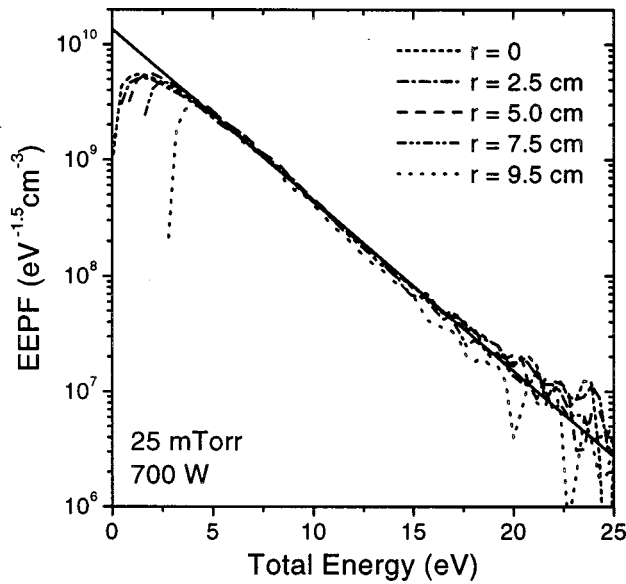


FIG. 3. Nonlocal EEPF in an O_2 plasma at 25 mTorr and 700 W for various radial positions. The solid line is a single-temperature Maxwellian fit to the data.

the coils for pressures below the nonlocality pR limit. This was attributed to the fact that the high-energy tail responsible for ionization is more localized than the medium and low energy electrons.²⁵ However, since the dissociation occurs at lower energies than ionization, the dissociation profile is expected to remain symmetric as long as the electrons are close to nonlocal. We have confirmed that the EEPF remains nonlocal at all conditions studied in this article. Further, the EEPF is well described by a single-temperature Maxwellian distribution in Figs. 3 and 4 (solid lines) and at all other plasma conditions studied in this article. The plasma density and electron temperature for various plasma conditions are

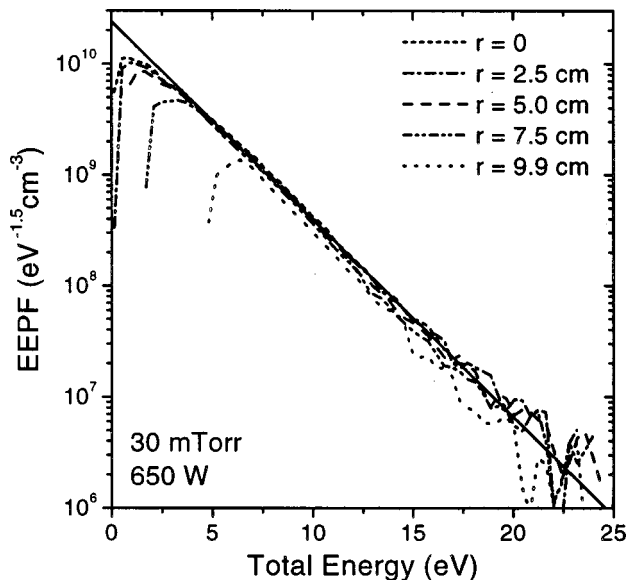


FIG. 4. Nonlocal EEPF in a N_2 plasma at 30 mTorr and 650 W for various radial positions. The solid line is a single-temperature Maxwellian fit to the data.

given in Table II. For nonlocal, approximately Maxwellian electrons, the spatial variation of the EEPF is given by

$$f_0(\epsilon, r, z) = f_0(\epsilon, 0, 0) e^{\phi(r, z)/T_e}, \quad (10)$$

where $\phi(r, z)$ is the plasma potential (in volts) with respect to the potential at the center of the discharge, and T_e is the electron temperature in volts. It should be noted that $\phi(r, z)$ is referenced to the potential at the center of the discharge, i.e., $\phi(0, 0) = 0$ and the plasma potential is negative at all other locations in the plasma.

We need to determine the variation of the plasma potential over the entire volume for the calculation of the volume generation rate of the radicals from Eq. (7). As stated before, we can measure the radial variation of the plasma potential using the Langmuir probe at a single axial location. We assume the dependence of the plasma potential on the dimensionless axial and radial position is the same, i.e.,

$$\phi(r, 0) = \phi(0, z) \quad \text{for } r/R = 2z/H. \quad (11)$$

Further, the plasma potential at any location in the plasma is approximated from

$$\phi(r, z) \approx \phi(r, 0) + \phi(0, z). \quad (12)$$

The assumptions made in Eqs. (11) and (12) are justified in the appendix. Based on these assumptions, we can determine $\phi(r, z)$ over the entire plasma volume based on our measurements of the radial variation of the plasma potential.

Using the aforementioned description of the variation of the parent number density in the volume, and the nonlocal behavior of electron in the plasma, Eq. (7) is simplified to

$$\frac{\gamma}{1 - 0.5\gamma} = \frac{8}{A_{\text{wall}} u_M} \left(\frac{n_{M_2}^{\text{wall}}}{n_M^{\text{wall}}} \right) k_{\text{diss}}(0, 0) n_e(0, 0) \times \int_{-H/2}^{H/2} \int_0^R e^{\phi(r, z)/T_e} 2\pi r dr dz. \quad (13)$$

The simplification from Eq. (7) to Eq. (13) is achieved because the electron temperature is invariant over the plasma volume as a result of the nonlocal, approximately-Maxwellian EEPF. The volume integral in Eq. (13) accounts for the effect of the variation of the plasma density on the generation term. The recombination coefficients calculated from Eq. (13) are given in Figs. 5 and 6. Figure 5 shows the mean value for γ_O is 0.17 ± 0.02 . Figure 6 shows the mean value for γ_N is 0.07 ± 0.02 . It should be noted that the ratio of the parent to radical number density varies by about a factor of 2, and the product $k_{\text{diss}}(0, 0) n_e(0, 0)$ varies by about a factor of 4 over the range of plasma conditions studied. However, the recombination coefficients estimated are within 30% of the mean. This suggests that the assumptions made in our model held well over the range of conditions studied, and the recombination coefficient can be treated as a constant over these conditions.

V. CONCLUDING REMARKS

We have estimated the recombination coefficients of O and N radicals using mass spectrometry and Langmuir probe measurements in an ICP. The measurements have been

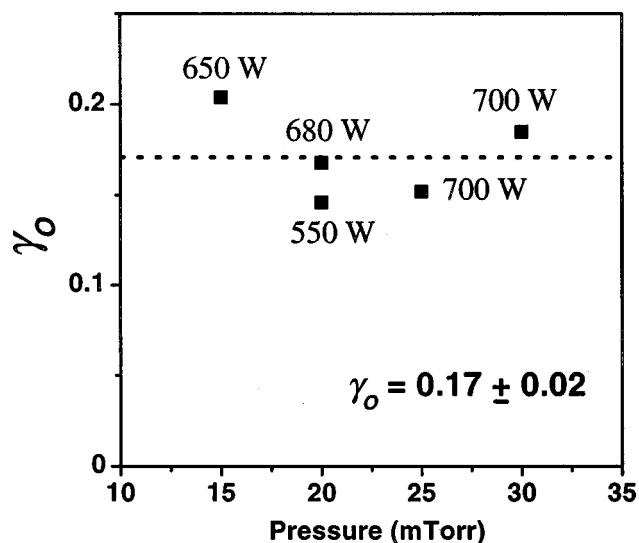


FIG. 5. Recombination coefficient of O radicals at various powers and pressures.

coupled with a plasma model describing the steady state balance between radical generation in the plasma and loss at the walls. The operation of the plasma at low pressures (10–30 mTorr) facilitates the use of a relatively simple model as the gradients in the neutral number densities are not appreciable and a global model for the plasma chemistry can be applied. The approximately nonlocal electrons at low pressures also facilitate the calculation of the volume generation rate of the radicals.

Our estimate of O recombination coefficient of 0.17 on stainless steel lies within the range 0.1–1 estimated by Granier *et al.*⁶ Booth *et al.*⁷ estimated the O recombination coefficient on stainless steel to be approximately 0.5. However, these authors stated that the precision of that measurement was insufficient for a quantitative determination.⁷ It should be noted that the measurements reported in this article

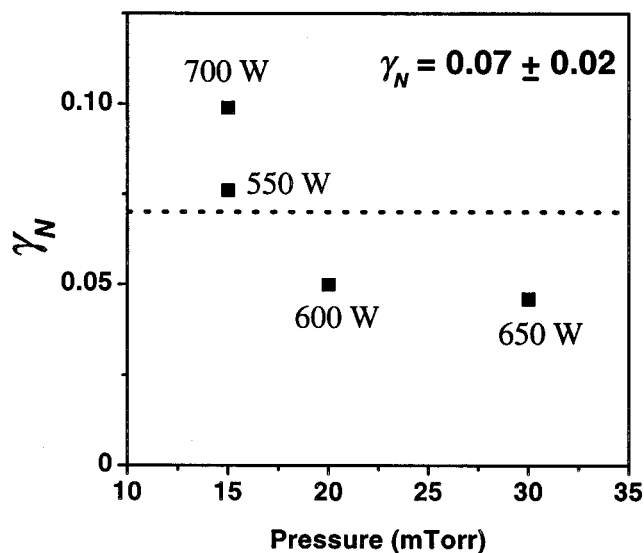


FIG. 6. Recombination coefficient of N radicals at various powers and pressures.

are made in the presence of ion bombardment from the plasma. The recombination coefficient of the radicals may be different in the absence of ion bombardment, for example, in the afterglow of a discharge. The largest uncertainty in our measurements is expected to arise due to the gradients in the neutral temperature in the plasma. A higher neutral temperature in the bulk plasma will lead to a lower total neutral number density in the discharge. This will lead to an overestimation of the rate of generation of the radicals by our model. Consequently, the recombination coefficient will be overestimated. Hebner²⁶ measured the neutral gas temperature profile in an Ar ICP. The neutral temperature in the center of the discharge was found to be 900 K for a neutral gas temperature of 600 K near the plasma boundary in that work. We estimate that for a similar temperature profile in our system, the recombination coefficient will be overestimated by about 30%.

These relatively large recombination coefficients of O and N radicals on stainless steel decrease the degree of dissociation in stainless steel reactors significantly as wall reactions play a dominant role in low-pressure discharges. Thus, the nature of the wall materials can be an important parameter in the design of plasma equipment for semiconductor processing. The surface reactions are often not well understood in these discharges. It should be noted that surface recombination coefficients are known to be relatively sensitive to changes in wall surface composition. The history of wall exposure to reactive gases could play a significant role in altering the surface recombination coefficient, even though the walls may nominally have the same composition. We note that measurements of the plasma composition (neutral and ionic) and the EEDF are key to understanding plasma chemistry. Finally, the combination of the experimental measurements and modeling studies can provide strong insights into the fundamental processes in these discharges.

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APPENDIX: ESTIMATION OF PLASMA POTENTIAL VARIATION OVER THE ENTIRE PLASMA VOLUME

We need to estimate the variation of the plasma potential over the entire plasma volume [$\phi(r, z)$] based on the radial variation of the plasma potential measured by the Langmuir probe. Consider the diffusion solution of the variation of the plasma density in a cylindrical volume obtained in terms of eigenfunctions,²⁷

$$n_e(r, z) = n_e(0, 0) J_0 \left(\frac{2.405r}{R} \right) \cos \left(\frac{\pi z}{H} \right), \quad (\text{A1})$$

where $n_e(0,0)$ is the peak plasma density and J_0 is the zero-order Bessel function. The Eq. (A1) can be approximated by

$$n_e(r,z) \approx n_e(0,0) \cos\left(\frac{\pi r}{2R}\right) \cos\left(\frac{\pi z}{H}\right), \quad (\text{A2})$$

since the behavior of the zero-order Bessel function in Eq. (A1) over $0 < r < R$ is quite similar to the cosine function. Integrating Eq. (10) over the energy domain, we get

$$n_e(r,z) = n_e(0,0) e^{\phi(r,z)/T_e}. \quad (\text{A3})$$

Comparing Eqs. (A2) and (A3),

$$\frac{\phi(r,z)}{T_e} \approx \ln \cos\left(\frac{\pi r}{2R}\right) + \ln \cos\left(\frac{\pi z}{H}\right). \quad (\text{A4})$$

It can be seen from Eq. (A4) that the functional dependence of the plasma potential on the dimensionless axial ($2z/H$) and radial (r/R) position is approximately the same. Further, the radial variation of the plasma potential at $z=0$ and at the location of our measurement ($z = -2.5$ cm) is the same. The above analysis supports the assumptions made in Eqs. (11) and (12). Further, the stated estimate of the plasma potential profile agrees within 10% with predictions of a fluid model simulation for this plasma geometry in an oxygen discharge at 15 mTorr.²⁸ It should be noted that while the diffusion solution might not be strictly valid for the conditions in our plasma, the eigenfunction separation of the radial and axial position dependence is expected to hold for a more accurate solution. Further, since the model used to calculate the dissociation utilizes integration over the volume, the final result is not very sensitive to spatial profiles used in Eqs. (11) and (12).

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